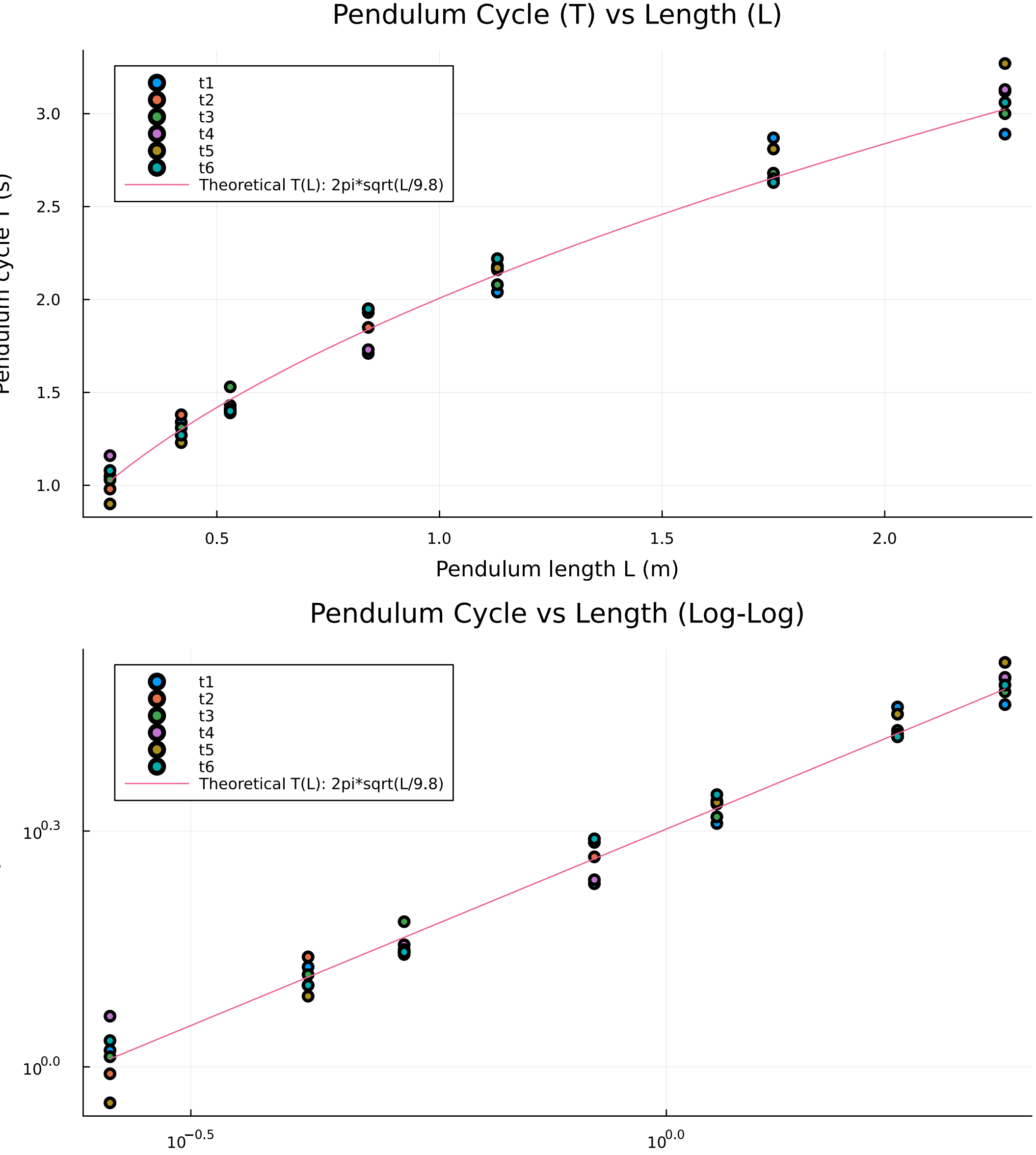


Julia code for creating the graphs

```
#note that there is an ~8ms slippage
#because of 120hz of my phone screen and the 1ms accuracy
#capturing a video of the video also has sync problems and is unrealistic,
#using the original clips is preferred.
using CSV, DataFrames, Plots
df = CSV.read("labwork_1/exp1.csv", DataFrame)
x = parse.(Float64, string.(df[:, 1]))
#the approximate length of the bottle of water is 16cm or 0.16m
#we measured from the bottom, to measure from the center mass it would be L-0.08m
println(x)
x=x.-0.08
println(x)
y = parse.(Float64, string.(Matrix(df[:, 2:end])))

p = scatter(x, y,
    label=["t1" "t2" "t3" "t4" "t5" "t6"],
    xlabel="Pendulum length L (m)",
    ylabel="Pendulum cycle T (s)",
    title="Pendulum Cycle (T) vs Length (L)")

f(L) = 2pi*sqrt(L/9.8)

plot!(p, f, minimum(x), maximum(x), label="Theoretical T(L): 2pi*sqrt(L/9.8)")
savefig(p, "pendulum_exp1.pdf")

p_log = scatter(x, y,
    label=["t1" "t2" "t3" "t4" "t5" "t6"],
    title="Pendulum Cycle vs Length (Log-Log)",
    xlabel="Pendulum length L (m)",
    ylabel="Pendulum cycle T (s)",
    xscale=:log10,
    yscale=:log10)
plot!(p_log, f, minimum(x), maximum(x), label="Theoretical T(L): 2pi*sqrt(L/9.8)")
savefig(p_log, "pendulum_exp1_log.pdf")
```